

Soft sensor: tray temperature → composition inference (Tier B)

Goal. Infer a distillation tray's liquid composition from its temperature and the column pressure alone — no tray-level GC — by inverting the vapor-liquid equilibrium relationship.

Most distillation columns have a temperature (and often pressure) transmitter on many trays; almost none have a composition analyzer on every tray — at best there's a GC on the overhead/bottoms product, if that. This is the canonical **Tier B** soft-sensor problem (docs/measurement_tiers.md): T and P are measured, composition is not, so composition has to be backed out by *inverting* the same equilibrium relationship notebooks 1–3 evaluated *directly*.

The catch, and why this is genuinely Tier B and not Tier A in disguise: a tray temperature and pressure alone only pin down composition if the system effectively has one composition degree of freedom — i.e. it's (pseudo-)binary. Here we model a depropanizer tray as a binary propane/n-butane "key component" system (a standard simplification when one light key and one heavy key dominate and everything else is a minor impurity). Given T and P, there is exactly one liquid composition consistent with the tray being at its bubble point, and we solve for it by bisection.

Measurements assumed available: tray temperature (T), column pressure (P), both continuous. **Inferred / computed by this soft sensor:** tray liquid composition (propane mole fraction), by inverting the bubble-point relationship. **Tier:** B — local inversion, not a direct calculation. If more than two components are present in non-trivial amounts, T and P no longer pin down a unique composition on their own and this degrades toward Tier C — you'd need either a fixed assumed profile for the non-key components or additional measurements.

This report reuses Clapeyron.jl's `bubble_temperature` (the forward direction) and inverts it numerically for composition by bisection, since Clapeyron doesn't offer that inverse directly for a general system, evaluated at a fixed operating point (the interactive notebook version lets you adjust tray T and column P with sliders).

```
function infer_light_key_fraction(model, P, T_target; tol=1e-4, maxiter=60)
    # bisection: T(x_propane) is monotonic decreasing, so this is well-posed
    ...
end
```

3. Soft sensor output: inferred tray composition

Tray temperature: **38.0 °C**, column pressure: **8.0 bar**.

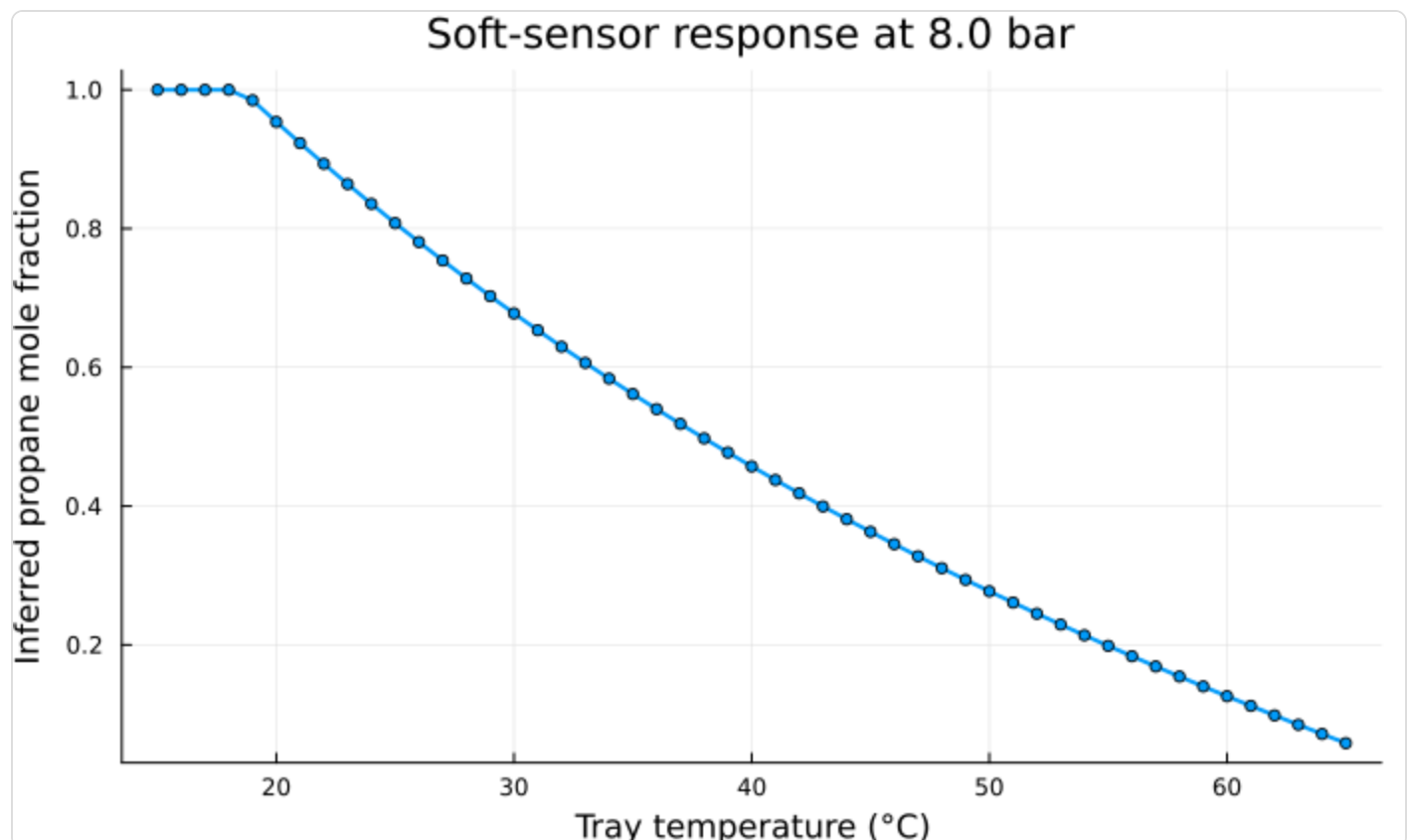
Inferred tray liquid composition (soft-sensor output): propane ≈ 0.498 , butane ≈ 0.502 (mole fractions), consistent with a bubble point of 38.0 °C at 8.0 bar.

No GC touched this tray — the composition comes entirely from inverting the same thermodynamics notebook 1 used in the forward direction.

Inferred propane mole fraction ≈ 0.498 at 38.0 °C / 8.0 bar

4. Sensitivity: inferred composition vs tray temperature

Sweep tray temperature at fixed column pressure and plot the inferred propane fraction — this is the soft sensor's whole operating curve. It should be smooth and monotonic (lower tray temperature $\Rightarrow$ more of the lighter, more volatile propane); a kink or discontinuity here would flag a numerical problem with the inversion, not a physical one.



5. Next steps toward a deployable soft sensor

- Validate the binary key-component assumption against an occasional multi-component tray sample (spot GC or manual sample) — if non-key components turn out to be significant, the inferred composition will be biased even though the inversion itself is exact for the assumed system.
- Apply the same inversion at every instrumented tray to build a full composition profile, and cross-check consistency (profile should be monotonic tray-to-tray in a well-behaved column) as a soft sanity check on both the model and the tray instrumentation.
- If a real tray has meaningful amounts of a third component, either fold it into the "heavy key" lump (as done here) and periodically re-validate that lump's effective properties, or add a second measurement to resolve the extra degree of freedom — otherwise this is a Tier C problem, not Tier B.
- Package `infer_light_key_fraction` into `src/` once a second Tier B notebook needs the same inversion pattern.

Generated from notebooks/04_tray_temperature_composition_soft_sensor.jl (Clapeyron.jl PR EOS) — caxelrud/Soft-sensor